Practical No. 02

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Roll No.:

Title - Write an Arduino/ Raspberry pi program for interfacing with PIR sensor Experiment.

CODE –

```
// Define pin numbers
const int pirPin = 2;    // PIR sensor input pin
const int ledPin = 13;   // LED output pin (built-in on many Arduino boards)
void setup() {  pinMode(pirPin, INPUT);    // Set PIR pin as input
  pinMode(ledPin, OUTPUT);   // Set LED pin as output
Serial.begin(9600);        // Initialize serial communication for debugging }
void loop() {  int pirState = digitalRead(pirPin);  // Read PIR sensor state
  if (pirState == HIGH) {              // If motion is detected
    digitalWrite(ledPin, HIGH);        // Turn on LED
    Serial.println("Motion detected!");
  } else {
    digitalWrite(ledPin, LOW);         // Turn off LED
    Serial.println("No motion");  }
delay(1000);  // Wait for a second before rechecking }
```

OUTPUT –

